

Machine Learning for Personalized Music Recommendation Based on User Listening Preferences

1. Title

Machine Learning for Personalized Music Recommendation Based on User Listening Preferences

2. Abstract

Music streaming platforms provide users with access to very large collections of songs, making it increasingly difficult for users to discover music that matches their interests. This project proposes the investigation of machine learning techniques for recommending music based on users' observed listening preferences and historical listening behaviour. The research will examine whether patterns in user-song interactions can be used to generate personalized recommendations.

The project will begin by exploring a publicly available music dataset to determine what information is available about users, songs, artists, and listening interactions. Based on the characteristics of the selected dataset, the team will define a precise recommendation task and identify appropriate machine learning approaches. A simple baseline method will first be established, followed by at least two substantive machine learning approaches suitable for the problem. The approaches will be evaluated using a consistent train, validation, and test methodology and recommendation-focused evaluation metrics such as Precision@K, Recall@K, or NDCG@K, where appropriate.

The expected contribution is a reproducible proof-of-concept demonstrating how machine learning can be applied to observed listening behaviour to produce personalized music recommendations, while also examining limitations such as sparse user histories, popularity bias, and recommendation errors.

3. Research Background

Music streaming services provide users with access to extremely large music libraries. While this gives users a wide range of choices, the size of these libraries can also make it difficult to discover music that is relevant to an individual listener. Recommendation systems address this problem by attempting to identify items that may be relevant to users based on information such as previous interactions and listening behaviour.

This project will investigate music recommendation using observed user listening preferences. Rather than attempting to determine whether a user personally "likes" a song, the project will focus on measurable behaviour contained within the dataset, such as previous interactions between users and songs. This distinction is important because listening behaviour provides an observable signal that can be used for machine learning without assuming that every recorded interaction represents an explicit rating or personal opinion.

Research question: How effectively can machine learning techniques use observed user listening behaviour to generate personalized music recommendations?

The project will first investigate the available data before determining the precise recommendation task. The team will examine factors such as the number of users and songs, available interaction information, the distribution of listening behaviour, and the sparsity of user-song interactions.

Existing recommendation approaches can use different types of information to identify potentially relevant items, including similarities between users, similarities between items, or patterns within historical interactions. Performance can be affected by limited listening histories, highly popular songs receiving disproportionate attention, and insufficient information about new users or songs.

Therefore, this project will not assume in advance that one particular machine learning technique will be the appropriate solution. Instead, the team will explore the dataset, define a measurable recommendation task, and then select suitable approaches that can be fairly compared. This will allow the project to investigate whether machine learning can generate recommendations and where the approaches have limitations.

4. Methodology

Data

The project will use a publicly available music dataset containing information about users' listening behaviour and interactions with music. Candidate datasets will be examined during the initial stage, with the final dataset selected based on relevance to the research question, available user-song interaction information, dataset size, documentation, licensing, and suitability for techniques covered in the course.

The team will investigate the selected dataset before defining the final recommendation task. Data preprocessing may include removing invalid or duplicate records, handling missing values where appropriate, examining interaction distributions, and transforming the data into a format suitable for machine learning.

Methods

The project will investigate multiple machine learning approaches suitable for the final recommendation task. The exact approaches will be selected after examining the dataset and defining the problem.

The project will include a simple task-appropriate baseline, at least two substantive machine learning approaches, appropriate preprocessing and feature construction, training and validation procedures, and model comparison using the same evaluation framework where appropriate.

Potential approaches may include methods for identifying similarities between users or items, but the final selection will depend on the characteristics of the dataset and the defined research problem. The team will also explain the theory of the selected approaches and how each method uses available information to generate recommendations.

Experiments and Evaluation

The dataset will be divided into training, validation, and test data using a methodology appropriate for the recommendation task. Training data will be used to develop models, validation data for model selection and parameter tuning, and the final test set for evaluation.

The experiments will compare machine learning approaches against the baseline under consistent conditions. The team will record preprocessing procedures, data splits, random seeds, model parameters, and experimental results to support reproducibility.

The project will also investigate model errors and limitations. Depending on the final dataset, this may include examining performance for users with different amounts of listening history or investigating whether recommendations are strongly influenced by popular songs.

Evaluation Metrics

Because the project focuses on recommendation rather than simply predicting a numerical value, recommendation-ranking metrics will be considered: Precision@K, which measures relevant recommendations among the top K; Recall@K, which measures relevant items successfully recommended within the top K; and NDCG@K, which evaluates ranking quality while giving greater importance to relevant items near the top. The final primary metric will be selected after the recommendation task and evaluation procedure have been finalized.

Comparison

The project will compare selected machine learning approaches against a simple baseline. The baseline will establish how well a straightforward recommendation strategy performs without the personalization provided by more substantive approaches. All approaches will be evaluated using the same appropriate data split and evaluation methodology wherever possible.

5. Significance

The project will contribute a reproducible proof-of-concept demonstrating how machine learning can be applied to user listening behaviour to address the problem of music discovery.

The results could be relevant to developers and researchers interested in recommendation systems, particularly those investigating how historical user-item interactions can be used to generate personalized recommendations. The project may also demonstrate practical challenges including sparse interaction data, popularity effects, and limited information about users or songs.

Rather than focusing only on producing recommendations, the project will examine how different approaches perform under a controlled evaluation process. This can provide insight into which characteristics of the data affect recommendation performance and where the approaches have limitations.

The findings could provide a foundation for future work involving additional information such as song characteristics, artist information, genre information, or more advanced recommendation techniques.

6. SMART Timeline

Sprint	Time Period	Objectives	Deliverable
1 — Problem & Data Investigation	Sept. 18 – Oct. 2	Investigate potential music datasets, examine available user/song information, identify a feasible recommendation problem, and define the research question.	Selected dataset, research problem, research question, initial data analysis

2 — Task & Baseline	Oct. 3 – Oct. 19	Define the recommendation task, establish train/validation/test methodology, determine evaluation metrics, implement an appropriate baseline, and establish initial results.	Evaluation procedure, baseline implementation, initial results
3 — ML Approach Investigation	Oct. 20 – Oct. 30	Investigate suitable ML approaches, select at least two substantive approaches, implement initial versions, and begin validation experiments.	Working ML approaches and initial comparison
4 — Final Experiments & Analysis	Oct. 31 – Nov. 20	Tune models using validation data, conduct controlled experiments, compare approaches against the baseline, analyze errors and limitations, and perform relevant robustness or sensitivity analysis.	Final experimental results, comparison, error analysis
5 — Finalization & Presentation	Nov. 21 – Dec. 4	Finalize the reproducible pipeline, documentation, report, presentation, and demonstration. Prepare for individual reflection/oral defense.	Final repository, report, presentation, demonstration

7. References

For the final submission, this section will contain the actual academic papers and dataset documentation used by the team. The proposal template permits APA or IEEE citation style. References will be added after the team selects and verifies the dataset and supporting research sources.